

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460022.7651 +/- 0.0042 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.04 +/- 0.053
Transit depth (Rp/Rs)^2	0.16 +/- 0.42 [%]
Orbital Inclination (inc)	84.71 +/- 1.74
Airmass coefficient 1 (a1)	0.89 +/- 0.17
Airmass coefficient 2 (a2)	0.09 +/- 0.24
Scatter in the residuals of the lightcurve fit is	18.67 %
Best Comparison Star	#2 - [445, 121]
Optimal Aperture	1.73
Optimal Annulus	6.13
Transit Duration (day)	0.026 +/- 0.031

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.04 ± 0.053 vs 0.1406 (deviation 1.9σ)
Duration fit vs published	0.026 d vs 0.0975 d (deviation 2.31σ)
Mid-time O-C	-5.8 min
Depth SNR	0.8
Residual scatter	18.67 %

Light curve

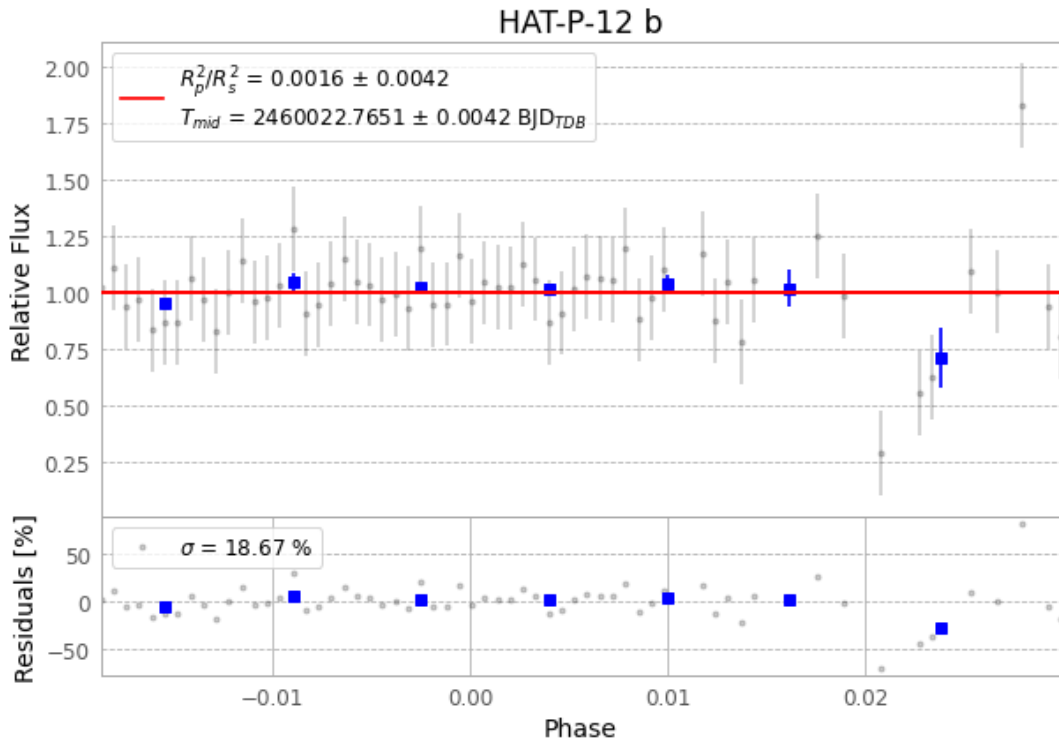


Figure 1 — FinalLightCurve_HAT-P-12 b_2023-03-18.png (SHA-256 1c187d57c341d35f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [517, 41], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ec9c5b37a1d3a8e5...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

74 FITS frames (49 MB), HATP-12230319044815.FITS → HATP-12230319083315.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-12-230319.log (4c7154d2cc4a...), FinalLightCurve_HAT-P-12 b_2023-03-18.png (1c187d57c341...), FinalLightCurve_HAT-P-12 b_2023-03-18.csv (bde392f3a6d1...)

Compute resources

Wall time 200.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 07:19Z.